Database Management

(ENCS205)

K.R. MANGALAM UNIVERSITY

Bachelor of Technology

In

Computer Science and Engineering

Submitted by:

Varun Chauhan 2401010276

Submitted to:

MS. MANSI KAJAL

School of Engineering & Technology

K.R. MANGALAM UNIVERSITY

Sohna, Haryana 122103, India

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ASSIGNMENT 2

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
▷Run
CREATE DATABASE LAB2;
▷Run
USE LAB2;

▷Run | Select
CREATE TABLE CLASS (
  id INT PRIMARY KEY,
  name VARCHAR(50)
);

▷Run
INSERT INTO CLASS VALUES (1, 'Varun');
▷Run
INSERT INTO CLASS VALUES (2, 'Jony');
▷Run
INSERT INTO CLASS VALUES (4, 'Bhawana');
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```
CREATE TABLE EMPLOYEES (
  Employee_Id INT PRIMARY KEY,
  First_Name VARCHAR(50),
  Last_Name VARCHAR(50),
  Email VARCHAR(100),
  Phone_Number VARCHAR(20),
  Hire_Date DATE,
  Job_Id VARCHAR(20),
  Salary DECIMAL(10,2),
  Commission_Pct DECIMAL(5,2),
  Manager_Id INT,
  Department_Id INT
);
```

3. How do you create a table named Client_master with the specified attributes?

```
▷Run | Select
CREATE TABLE Client_master (
  ClientNO VARCHAR(10) PRIMARY KEY,
  Name VARCHAR(50),
  Address1 VARCHAR(100),
  City VARCHAR(50),
  Pincode INT,
  State VARCHAR(50),
  Bal_due DECIMAL(10,2)
);
```

4. What SQL statement would you use to rename the table Client_master to Client12?

```
ALTER TABLE Client_master
RENAME TO Client12;
```

5. How do you display the bal_due column heading as “BALANCE”?

```
SELECT Bal_due AS "BALANCE"
FROM Client12;
```

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;
- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;
- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;

- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

```
INSERT INTO CLASS VALUES (5, 'Priyanka');
▷Run
COMMIT;
▷Run
UPDATE CLASS SET name = 'Vanshika' WHERE id = 5;
▷Run
SAVEPOINT A;
▷Run
INSERT INTO CLASS VALUES (6, 'Anjali');
▷Run
SAVEPOINT B;
▷Run
INSERT INTO CLASS VALUES (7, 'Mrinal');
▷Run
SAVEPOINT C;
▷Run | +Tab | JSON
SELECT * FROM CLASS;
▷Run
ROLLBACK TO B;
▷Run | +Tab | JSON
SELECT * FROM CLASS;
▷Run
ROLLBACK TO A;
▷Run | +Tab | JSON
SELECT * FROM CLASS;
```

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
INSERT INTO EMPLOYEES VALUES
(101, 'Varun',    'Chauhan',   'varun@gmail.com',    '1234567890', '2003-06-17', 'AD_PRES', 24000, NULL, NULL, 90),
(102, 'Jony',     'Gautam','jony@gmail.com',     '1234567891', '2005-09-21', 'AD_VP',   17000, NULL, 101,  90),
(103, 'Bhawana',  'Jindal',   'bhawana@gmail.com',  '1234567892', '2001-01-13', 'AD_VP',   17000, NULL, 101,  90),
(104, 'Priyanka', 'Kumar',    'priyanka@gmail.com', '1234567893', '2006-01-03', 'IT_PROG',  9000, NULL, 103,  60),
(105, 'Vanshika', 'Dey',      'vanshika@gmail.com', '1234567894', '2007-05-21', 'IT_PROG',  6000, NULL, 103,  60),
(106, 'Anjali',   'Chawla',   'anjali@gmail.com',   '1234567895', '2005-06-25', 'IT_PROG',  4800, NULL, 103,  60),
(107, 'Mrinal',   NULL,       'mrinal@gmail.com',   '1234567896', '2006-02-05', 'IT_PROG',  4800, NULL, 103,  70),
(108, 'Aryan',    'Bisht',    'aryan@gmail.com',    '1234567897', '2007-02-07', 'IT_PROG',  4200, NULL, 103,  80);
▷Run | +Tab | JSON
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Search Results Export

Cost: 10ms 1 Total 8

Employee_Id int	First_Name varchar(50)	Last_Name varchar(50)	Salary decimal(10,2)
101	Varun	Chauhan	24000.00
102	Jony	Gautam	17000.00
103	Bhawana	Jindal	17000.00
104	Priyanka	Kumar	9000.00
105	Vanshika	Dey	6000.00
106	Anjali	Chawla	4800.00
107	Mrinal	(NULL)	4800.00
108	Aryan	Bisht	4200.00

2. List out the employees who work under manager 101.

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 101;   5ms
```

Employee_I int	First_Name varchar(50)	Last_Name varchar(50)	Email varchar(100)	Ph
102	Jony	Gautam	jony@gmail.com	12
103	Bhawana	Jindal	bhawana@gmail.com	12

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;   5ms
```

First_Name varchar(50)	Last_Name varchar(50)
Varun	Chauhan
Jony	Gautam
Bhawana	Jindal
Priyanka	Kumar
Vanshika	Dey
Anjali	Chawla
Mrinal	(NULL)

4. List out the employees whose last name is ‘chawla’.

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Chawla';
```

Employee_I int	First_Name varchar(50)	Last_Name varchar(50)	Email varchar(100)	Phon var
106	Anjali	Chawla	anjali@gmail.com	1234

5. Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60, 70, 80);   5ms
```

First_Name varchar(50)	Last_Name varchar(50)
Priyanka	Kumar
Vanshika	Dey
Anjali	Chawla
Mrinal	(NULL)
Aryan	Bisht

6. Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;   25ms
```

Manager_Id int
(NULL)
101
103

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

```
INSERT INTO Client12 VALUES
('C201', 'Varun',    'Street 1', 'Delhi',   110001, 'Delhi',        6000),
('C202', 'Jony',     'Street 2', 'Mumbai',  400001, 'Maharashtra',  3000),
('C203', 'Bhawana',  'Street 3', 'Chennai', 600001, 'Tamil Nadu',   8000),
('C204', 'Priyanka', 'Street 4', 'Delhi',   110002, 'Delhi',        2000),
('C205', 'Anjali',   'Street 5', 'Mumbai',  400002, 'Maharashtra',  7000);
```

2. Find the names of clients whose bal_due > 5000.

```
SELECT Name, Bal_due
FROM Client12
WHERE Bal_due > 5000;   6ms
```

Name varchar(50	Bal_due decimal(10,2
Varun	6000.00
Bhawana	8000.00
Anjali	7000.00

3. Change the bal_due of ClientNO "C203" to Rs. 5100.

```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNO = 'C203';
```

```
SELECT *
FROM Client12
WHERE ClientNO = 'C203';
```

Name varchar(50	Bal_due decimal(10,2
Varun	6000.00
Bhawana	8000.00
Anjali	7000.00

ClientN varchar(10	Name varchar(50	Address1 varchar(100	City varchar(50	Pincode int	State varchar(50	Bal_due decimal(10,2
C203	Bhawana	Street 3	Chennai	600001	Tamil Nadu	5100.00